Applied Machine Learning for Business

Course Project Instructions

Plaksha University | Spring 2026 Instructor: Nikhil George

Teams and Logistics

Teams of 2. Your interim submission on February 17 identifies your team; no separate registration needed. Both members must contribute to every phase. Your final report includes a contribution statement.

Deliverable	Due	Weight
Interim Written Submission	Feb 17, 11:59 PM IST	10%
Mid-Project Check-In	Mar 17–21 (in class, written)	Ungraded
Final Presentation	Apr 23 – May 4 (in class)	15%
Final Report + GitHub Repository	Apr 23, 11:59 PM IST	15%
Total		**40%**

The report and repository are due **April 23 for all teams**, regardless of presentation slot. Presentations are scheduled by the instructor.

What a Project Is

A project has two layers:

Layer 1: Empirical. A real dataset. Real feature engineering, model building, and evaluation. This is non-negotiable—there is no project without data and working code.

Layer 2: Economic. A constructed business scenario built on top of the empirical work. You are not the firm. You will not have their cost structure, their customer base, or the ability to run experiments. Instead, you do what we do in lecture: assume reasonable numbers, state them, and derive the insight that follows.

In the credit scoring lecture, we assumed G = 60K and L = 500K. We did not know the bank's actual numbers. But the insight—that nearly all the value of a better model comes from avoided losses, not found gains—holds across a wide range of assumptions. That is what your project should aim for: an insight that survives reasonable perturbation of the assumptions.

Choosing the right problem is a large part of the work. A well-chosen problem—the right dataset, the right decision-maker, the right framing—makes the rest of the project flow naturally. A poorly chosen one creates friction at every stage. The ability to look at a dataset and see the business question hiding inside it is a skill this course develops and one the grading rewards. Do not rush this step.

The structure of a good project:

1. **Start from a real dataset.** Build models. This is the foundation.

2. **Identify a decision-maker.** Who would use this model's output, and for what decision?
3. **Define the economics.** What does each type of correct and incorrect prediction cost or save? Assume numbers. Defend them.
4. **Connect model output to the decision.** What threshold, ranking, or operating point would the decision-maker use? Evaluate against this—not just accuracy.
5. **Test sensitivity.** Since you assumed the economics, show what happens when they change. Which assumptions matter and which don't?
6. **Articulate limitations and the next investment.** What would you need—data, access, experiments, infrastructure—to move from this analysis to a deployed system? What can't be known without being the firm?

Example: Vehicle Sound Classification for Toll Pricing

You find a labeled dataset of vehicle audio recordings from Indian roads—clips classified as car, auto-rickshaw, two-wheeler, bus, truck, LCV. The ML task is audio classification: extract spectral features (MFCCs, spectral centroid, roll-off), train a multi-class classifier, evaluate per-class performance. That work is necessary. It is not yet a project.

The project begins when you identify the decision-maker: a toll plaza operator on an NHAI highway. Audio sensors are cheap, robust to dust and low visibility, and could supplement or replace camera-based classification. Toll rates are public and differ by vehicle class—cars pay ₹100, LCVs ₹150, buses ₹200, trucks ₹350. Every misclassification has a cost. Classify a truck as a car: you lose ₹250 in toll revenue. Classify a car as a truck: the driver disputes, costing an assumed ₹50 in processing and goodwill. This gives you a **confusion cost matrix**—a different cost for every pair of classes—not a single accuracy number.

Now multiply your model's confusion matrix by this cost matrix at a given daily traffic volume. You can ask real questions: which misclassification pairs drive the most cost? Does a model with 3% lower overall accuracy but fewer truck-car confusions actually save more money? What happens during harvest season when truck volume doubles? If a two-wheeler is misclassified, does it matter economically at all?

Notice that the dataset includes a broad set of vehicle classes—and the model will not classify all of them well. That is a feature, not a bug. If the classifier reliably confuses LCVs and trucks, the economically rational response may be to collapse them into a single toll tier: you sacrifice some revenue differentiation to eliminate a costly source of misclassification. Choosing the optimal number of classes is itself an economic decision—one that depends on the confusion rates your model produces and the toll rate differences between classes. The model tells you what it can and cannot distinguish; the economics tell you whether the distinction is worth attempting.

The limitations section writes itself: audio alone can't distinguish vehicles with similar engine profiles (e.g., a loaded vs. unloaded LCV). A paired audio-visual dataset collected at actual toll plazas would resolve this, but requires field deployment and labeling infrastructure. That is the next investment.

Deliverables

Interim Submission (10%) — Feb 17

PDF, 2–4 pages. This is evidence of work already done, not a proposal of intentions.

1. **Problem and Decision (half page).** The business problem, the decision-maker, and the action your analysis informs. State the assumptions you are making about costs, constraints, and objectives. "Predict which subscribers will cancel in the next 30 days so the retention team can prioritize 500 outreach slots" — not "predict churn."
2. **Data (half page).** Source, size, time period, key variables. Confirm access: include a sample (5–10 rows) or screenshot. If multiple sources, explain how they join.
3. **Summary Statistics (half page).** Distribution of the target variable, missing value rates, and at least one visualization that reveals something non-trivial about the data.
4. **Proposed Approach (half page).** Methods you plan to use and why. What features will you engineer? What models will you compare? What is one challenge you anticipate (class imbalance, temporal leakage, limited labels)? How will you evaluate in a way that reflects the business problem?

Mid-Project Check-In (Ungraded) — Mar 17–21

Written, 2 pages, submitted in class. This is a forcing function: it ensures you are on track and gives the instructor a chance to course-correct before the final push.

1. **Progress.** What have you completed since the interim? Specifically: data cleaning status, features engineered, models trained (if any), preliminary results.
2. **Preliminary Findings.** At least one concrete result—a baseline model's performance, a surprising pattern in the data, or an initial version of the economic analysis. Show a number, a table, or a figure. "We are still cleaning data" is not sufficient.
3. **Revised Plan.** What remains? Has the scope changed since the interim? What is the biggest open risk to completing the project?

The check-in is not graded, but failure to submit will result in a 5% penalty on your final project grade.

Final Deliverables (30%) — Presentation Apr 23–May 4; Report + Repo Apr 23

Presentation (15%). 15 minutes + 5 minutes Q&A. Both members present. Structure: business context (2–3 min), data and features (3–4 min), methodology (3–4 min), results translated into business impact (3–4 min), limitations and next investment (1–2 min). Minimize text on slides. A well-chosen chart communicates more than three slides of bullet points.

Report (10%). PDF, 8–12 pages excluding appendices. Must stand alone—a reader who missed the presentation should understand your full analysis. Sections: Introduction and business framing. Data description. Feature engineering (include features you tried that did not work). Methodology and model comparison. Results with business-relevant metrics. Economic analysis with sensitivity. Limitations and next investment. Contribution statement.

GitHub Repository (5%). Organized, reproducible code. README with project description, data access instructions, and repo structure. Notebooks/scripts numbered in execution order (e.g., `01_cleaning.ipynb`, `02_features.ipynb`, `03_modeling.ipynb`). Include `requirements.txt`. No API keys or credentials.

Grading

Interim Submission (10%)

Criterion	Weight
Problem clarity: specific, testable question tied to a decision with stated assumptions	25%
Data readiness: data in hand, loaded, described with evidence	30%
Initial exploration: meaningful statistics and at least one non-trivial observation	25%
Feasibility: realistic plan with awareness of at least one challenge	20%

Final Deliverables (30%)

Dimension	Weight	Excellent	Needs Improvement
Business context and scenario	30%	Well-chosen problem; clear decision-maker; well-defended assumptions; cost structure that generates non-trivial analysis	No decision identified; poor problem choice; economics absent or asserted without reasoning
ML and empirical work	20%	Thoughtful feature engineering; multiple models compared with justification; proper train/test discipline	Raw features only; single model; data leakage or flawed evaluation
Economic analysis and insight	30%	Model output connected to business outcome; sensitivity analysis on key assumptions; insight that would change a decision-maker's action	Accuracy reported with no business translation; no sensitivity analysis
Limitations and next investment	10%	Honest assessment of what cannot be known from this data; specific next steps with reasoning (data, experiments, infrastructure)	Generic "we need more data" with no specifics

Dimension	Weight	Excellent	Needs Improvement
Communication	10%	Clear presentation; polished report; well-organized and reproducible code	Confusing structure; missing deliverables; code not runnable

The ML work (20%) is necessary but it is the part you already know from prerequisites. The value-add of this course is everything around it: framing the problem, constructing the economics, extracting the insight, and knowing where the analysis ends and the next investment begins. The grading reflects this.

What to Avoid

This is not an exhaustive list, but these are common failure modes:

- Running five models on a competition dataset and reporting which has the best AUC is a homework problem, not a project. You may use competition datasets, but you must add the economic layer.
- A prediction task with no downstream action is a forecasting exercise, not a business application. Anchor your work in a decision someone would actually make differently based on your analysis.
- A focused analysis done well beats an ambitious one done halfway. You have ~10 weeks.
- Sending prompts to an LLM and displaying the output is not an ML project. The core must involve working with data: cleaning, feature engineering, training, evaluating.

Academic Integrity

You may use publicly available code, libraries, and pre-trained models—cite them in your report and README. Using AI tools for coding assistance and debugging is permitted, but you must understand and be able to explain every part of your submission. Copying another team's work, or submitting analysis you cannot reproduce, will be reported to the Academic Integrity Committee.

Dual submission: If you intend to submit this project (in whole or in part) for another course at Plaksha, you must obtain written permission from the instructor **before your first submission** (i.e., before the interim deadline). Any dual submission discovered after the fact without prior approval will be reported to the Academic Integrity Committee.

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